

Ashkan Arabi

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EDUCATION

Bachelor of Science, Computer Science

The University of Texas at El Paso

December 2025

GPA: 3.93

Certifications: AWS Certified Solutions Architect - Associate (April 2026)

EXPERIENCE

Capital One: Associate Software Engineer

February 2026 - present | New York, NY

- Built a **Golang** microservice, deployed using **AWS Lambda**, enabling internal services to read and write customer tax withholding data during migration from the legacy FIS Profile database.
- Wrote **Grafana K6** load tests simulating Discover user data migration into Capital One's customer DB.
- Added **OpenTelemetry** logging support across team microservices using Otelslog, cutting logging costs by **68%** by migrating from Splunk to Observe and improving incident response through log correlation.

Capital One: Software Engineering Intern

June 2025 - August 2025 | McLean, VA

- Built & deployed anomaly-detection pipeline for Capital One's card decisioning system, notifying SREs of malfunctioning ML models and reducing incident-response time **from 4+ hours to <15 mins**.
- Wrote highly optimized **Databricks SQL** & evaluated warehouse sizes to minimize pipeline cost & latency.
- Extended **Java**-based plugin to log custom **Apache Spark** data lineage events using a YAML specification.
- Applied **test-driven development** using **JUnit** and **Mockito** to ensure robust, maintainable code.

Texas Instruments: Information Technology Intern

May 2024 - August 2024 | Dallas, TX

- Deployed, tested, and monitored production **Docker** infrastructure for semiconductor wafer transporter robots, reducing projected fab operating costs by **~34%**.

UTEP Vision and Learning Lab: Undergraduate Research Assistant

January 2024 - May 2024 | El Paso, TX

- Built tone-changer by fine-tuning **HuggingFace** models to change normal tone to autistic.
- Implemented neural style-transfer model in **PyTorch** to explore usage for accent-changing.
- Increased evaluation script performance by **144x** reimplementing KNN in PyTorch to utilize GPU.

Temple University: Research Intern

June 2023 - July 2023 | Philadelphia, PA

- Published paper in **ACM MobiHoc** about using Wi-Fi CSI for hand gesture recognition on smartphones.
- Developed CNN model to classify 5 gestures in 6 scenarios using **TensorFlow** with **>90% accuracy**.

PROJECTS

Movie Rec Engine | FAISS, Redis, gRPC, PyTorch, FastAPI, Postgres, AWS

- Implemented production-grade two-stage (candidate-retrieval + ranking) **collaborative-filtering**-based movie recommender, deployed to **GPU-enabled AWS EC2**.
- Separated retrieval & ranking services for independent scaling, connected by **gRPC** for minimum overhead.
- Used **FAISS** for GPU-accelerated candidate retrieval & **pre-warmed, Redis + Postgres write-through cache** to minimize latency when fetching repeated embeddings.

Ashdocs | Go(Golang), Websocket, Automerge

- Built **multi-threaded** real-time collaborative editor in **Go**, capable of **930+ edits/s** on Apple M4.
- Integrated **Automerge CRDTs** for conflict-free merging of simultaneous edits with **eventual consistency**.
- Designed event pipeline using **goroutines** and **WebSockets** to broadcast per-document updates.
- Created **Pebble**-backed operation log w/ **periodic full snapshots** to cap operation replay time.

Newsbridge (news.ashkan.zone) | FastAPI, React, Postgres, JavaScript, Docker, Ollama, Gemini API

- Increased access to diverse news by creating an AI RSS news translator & orator with **55 GitHub stars**.
- Allowed instant playback using **async** programming to **concurrently generate** summary & play audio.